

SNEHA SONKUSARE

AI Engineer | LLM Inference & MLOps

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SUMMARY

AI Engineer with 2+ years designing, fine-tuning, and deploying LLM/SLM inference platforms, agentic AI systems, and RAG pipelines; open-source contributor to PyTorch (documentation). Specializes in parameter-efficient fine-tuning (LoRA, QLoRA, Unsloth), GPU-accelerated model serving on Kubernetes, and context engineering for production LLM applications. Builds end-to-end MLOps pipelines spanning observability, autoscaling, and AI governance (auth, rate limiting, drift monitoring) across AWS, GCP, and NVIDIA NIM, with current work in sovereign AI architecture for enterprise deployment.

SKILLS

Languages: Python, R, SQL, Rust

LLM Engineering & GenAI: LLMs/SLMs, RAG, Agentic AI & Multi-Agent Orchestration, Prompt & Context Engineering, Fine-Tuning (LoRA, QLoRA, Unsloth, PEFT), NVIDIA NIM, AWS Bedrock, Groq API

Inference & MLOps: Docker, Kubernetes (GKE), Model Serving & Inference Optimization, KEDA Autoscaling, MLflow (Experiment Tracking), Model Monitoring & Drift Detection, CI/CD (GitHub Actions)

AI Governance & Observability: Kong API Gateway (Auth, Rate Limiting, Routing), Prometheus/Grafana, Latency & Drift Monitoring, Responsible AI

Deep Learning: PyTorch, TensorFlow, CNNs, LSTMs, NLP

Cloud & Data Platforms: AWS (S3, Glue, SageMaker, QuickSight), GCP, Snowflake, Amazon Redshift, PostgreSQL, MySQL, Apache Spark, Airflow, dbt

Classical ML & Statistics: Classification, Regression, Clustering, XGBoost, LightGBM, SVMs, Hypothesis/A-B Testing, Bayesian Inference

Visualization: Tableau, Power BI, Seaborn, Matplotlib

EXPERIENCE

AI Engineer Intern | Hexaware Technologies Inc.

Jul 2026 – Present

Sovereign AI Architecture & Enterprise Deployment

Remote, USA

- Build and deploy enterprise RAG pipelines, Voice AI systems, and Co-pilot applications for sovereign AI architecture initiatives targeting enterprise deployment.
- Run SLM fine-tuning experiments (LoRA/QLoRA, Unsloth) and benchmark performance on domain tasks across NVIDIA NIM and AWS Bedrock.
- Develop Agentic AI proof-of-concepts, including multi-agent architectures, tool-use integrations, and context engineering strategies for reliable long-context reasoning.
- Contribute to frontend development for AI-facing user interfaces, prioritizing latency and usability.

Graduate Research Assistant | UMass Dartmouth School for Marine Science & Technology

Jan 2025 – Dec 2025

New Bedford, MA

- Performed EDA on 500K+ oceanographic records, engineered domain-specific features, and trained Random Forest and CNN models (PyTorch/TensorFlow) achieving 85% accuracy; tracked experiments via MLflow and containerized with Docker for reproducibility.
- Optimized model performance using cross-validation and precision/recall/F1 tuning; translated insights for non-technical teams to inform conservation decisions.
- Standardized metadata and data governance across 5+ systems, reducing inconsistencies by 25% and improving reproducibility.
- Built Apache Spark pipelines for sensor data processing and CI/CD workflows for automated testing and redeployment, improving data accuracy by 20% through collaboration with marine scientists.

Data Analyst | Hexaware Technologies Inc.

Sep 2021 – Jan 2024

Mumbai, India

- Collaborated with finance, operations, and product stakeholders to translate pain points into data questions, writing complex SQL queries across 5+ systems for a unified data layer, reducing errors by 30% and saving 10+ hours weekly.
- Detected anomalies in fraud and credit risk data using EDA and statistical hypothesis testing; applied Scikit-learn classification models improving decision accuracy by 25%, and maintained 8+ Power BI dashboards, cutting report generation time by 35%.
- Established 15+ KPIs across teams with senior stakeholders, enabling self-serve analytics that reduced ad-hoc requests by 20% and eliminated manual reporting rebuilds.
- Resolved 20+ pipeline failures in Agile teams with senior engineers, using Docker to reproduce bugs, boosting processing speed by 30% and reducing query runtime by 40%.

PROJECTS

InfraLLM | GitHub

May 2026 – May 2026

- Architected and deployed a production-grade, GPU-accelerated LLM inference platform on GKE, serving Gemma 2B, LLaMA 3 8B, and Mistral 7B via containerized workloads, with Kong API Gateway enforcing auth, rate limiting, and request routing for AI governance.
- Implemented MLOps observability with Prometheus/Grafana, KEDA event-driven autoscaling, and GitHub Actions CI/CD for zero-touch model deployments.

AI-Powered Learning Platform (Context Engineering) | Live App

May 2026 – Jun 2026

- Engineered a context-engineering pipeline that consolidated 7 sequential LLM API calls into a single structured prompt across LLaMA 3.3 70B (Groq API), cutting generation latency by 80% while powering 10+ tools (Mock Interview, Quiz Generator, AI Doubt Solver).
- Designed a 3-model fallback/routing chain for inference reliability and multilingual support across 12-chapter auto-generated courses; deployed on Streamlit.

Instruction Fine-Tuning Based Medical Chatbot | GitHub

Apr 2025 – May 2025

- Fine-tuned DeepSeek-R1 in PyTorch (LoRA/QLoRA via Unsloth) on medical Q&A and built a Neo4j-backed RAG pipeline for structured clinical retrieval, improving response accuracy by 30%.
- Developed a production-ready Flask REST API inference pipeline with real-time latency/drift monitoring, containerized for AWS deployment.

EDUCATION

University of Massachusetts Dartmouth

Jan 2024 – Dec 2025

Master of Science in Data Science | GPA: 3.73/4.00 (Relevant Coursework: Artificial Intelligence, Advanced Machine Learning, Big Data Analytics, Database Design)

Maharashtra Institute of Technology

Aug 2017 – Jul 2021

Bachelor of Technology – Electronics and Communication